Hints: Combinatorics – 1. Counting and bijections

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Stars and bars: a multiset of size k from $\{1,\dots,n\}$ is a sequence $m_1+\cdots+m_n=k$ of multiplicities; write m_1 stars, a bar, m_2 stars, a bar, and so on. Multinomial theorem: expand $(x_1+\cdots+x_m)^n$ as a sum over words of length n and group words by how many times each letter occurs.

Hint 2. Hockey stick $\sum_{m=k}^{n}\binom{m}{k}=\binom{n+1}{k+1}$: classify the $(k+1)$-subsets of $\{1,\dots,n+1\}$ by their largest element $m+1$. Lattice paths: a path to (a,b) is a word of a letters E and b letters N, determined by the positions of the E's.

Hint 3. Paths avoiding a point (c,d) with $0\leqslant c\leqslant a$, $0\leqslant d\leqslant b$: subtract those passing through it, which split uniquely into a path to (c,d) followed by a path from (c,d). Answer: $\binom{a+b}{a}-\binom{c+d}{c}\binom{a+b-c-d}{a-c}$.

Problem 2

Hint 1. Inclusion–exclusion: fix an element lying in exactly $m\geqslant 1$ of the sets and count how many times it is counted on the right-hand side; it is $\sum_{j\geqslant 1}(-1)^{j+1}\binom{m}{j}=1-(1-1)^m$.

Hint 2. Derangements: A_i = permutations fixing i, so $\left|\bigcap_{i\in I}A_i\right|=(n-|I|)!$; this gives $D_n=n!\sum_{j=0}^{n}(-1)^j/j!$ and $|D_n/n!-e^{-1}|\leqslant 1/(n+1)!$ (alternating series tail). Surjections $[n]\to[k]$: A_i = maps missing i, giving $\sum_j(-1)^j\binom{k}{j}(k-j)^n$. For $\varphi(n)$: A_p = multiples of p in $[1,n]$ for the primes $p\mid n$, and the sum factors as a product.

Hint 3. No fixed point among the first k positions: apply inclusion–exclusion only to $A_1,\dots,A_k$. The answer is $\sum_{j=0}^{k}(-1)^j\binom{k}{j}(n-j)!$.

Problem 3

Hint 1. For $k\geqslant 2$ the condition reads $f(f(k))=f(k-1)$. List all such functions for $n=4,5$ and look for a pattern. Let $a=f(1)$ be the minimum value and let p be the largest index with $f(p)=a$ (so $p<n$).

Hint 2. Show that f is strictly increasing on $\{p+1,\dots,n\}$: if $f(k)=f(k-1)$ for some $k\geqslant p+2$, then $f(k-1)=f(f(k))=f(f(k-1))=f(k-2)$, and this propagates down to $f(p+1)=f(p)$. Deduce from $f(f(k))=f(k-1)>a$ that $f(k)>p$ and then that $f(k)=k-1$ for $k\geqslant p+2$.

Hint 3. So $f=a$ on $[1,p]$, $f(p+1)=j$, $f(k)=k-1$ for $k\geqslant p+2$, and the remaining condition $f(j)=f(p)=a$ forces $a<j\leqslant p$. Conversely all such functions work. Count triples $1\leqslant a<j<p+1\leqslant n$: $|F|=\binom{n}{3}$.

Problem 4

Hint 1. Fix the number of parts k. What do the conditions say about the multiset $\{a_1,\dots,a_k\}$?

Hint 2. All parts are q or $q+1$ for some q. Writing $n=kq+r$ with $0\leqslant r<k$ shows that for each $k\in\{1,\dots,n\}$ there is exactly one such representation (r parts equal to $q+1$, the rest equal to q). The answer is n.

Problem 5

Hint 1. Each jump increases one coordinate by 1 (there are $3n$ jumps and the total increase is $3n$). Encode a path as a word in letters x, y, z. The condition $y \leqslant x \leqslant y+1$ says the letters x and y alternate $x, y, x, y, \ldots$; what does $z \leqslant y$ say?

Hint 2. Map x and y to an up-step $+1$ and z to a down-step -2. Check that the height after a prefix, $\#x + \#y - 2\#z$, is $\geqslant 0$ exactly when $\#z \leqslant \#y$, and that the x/y labels are recovered from the alternation. So paths correspond to sequences of $2n$ steps $+1$ and n steps -2 with all partial sums non-negative.

Hint 3. Count those by the cycle lemma: prepend one extra $+1$ step, so there are $2n+1$ steps $+1$ and n steps -2 with total $+1$; among the $3n+1$ cyclic rotations of any such sequence exactly one has all partial sums positive, giving $\frac{1}{3n+1}\binom{3n+1}{n}$. The answer is $\frac{1}{2n+1}\binom{3n}{n}$ (1, 3, 12, 55, ...).